

Stock Price Prediction Using LSTM Networks

- **Using Apple (AAPL) Stock Data with MACD and RSI Indicators**
- **Name:** Riva Mehta
- **Roll No:** 25B2137

Problem Statement

- **Objective**
Build a deep learning model that:
 1. Downloads stock data from Yahoo Finance
 2. Uses technical indicators
 3. Learns historical patterns
 4. Predicts the next 15 trading days
- **Stock Used**
 1. Apple (AAPL)
 2. January 2020 – January 2025

Data Collection

- **Data Source**
 1. Yahoo Finance
 2. Python Library: yfinance
- **Data Downloaded**
 1. Open
 2. High
 3. Low
 4. Close
 5. Volume
- **Feature Selected**
 1. Close Price



Technical Indicators

- **MACD (Moving Average Convergence Divergence)**

Measures:

1. Trend strength
2. Trend direction
3. Momentum

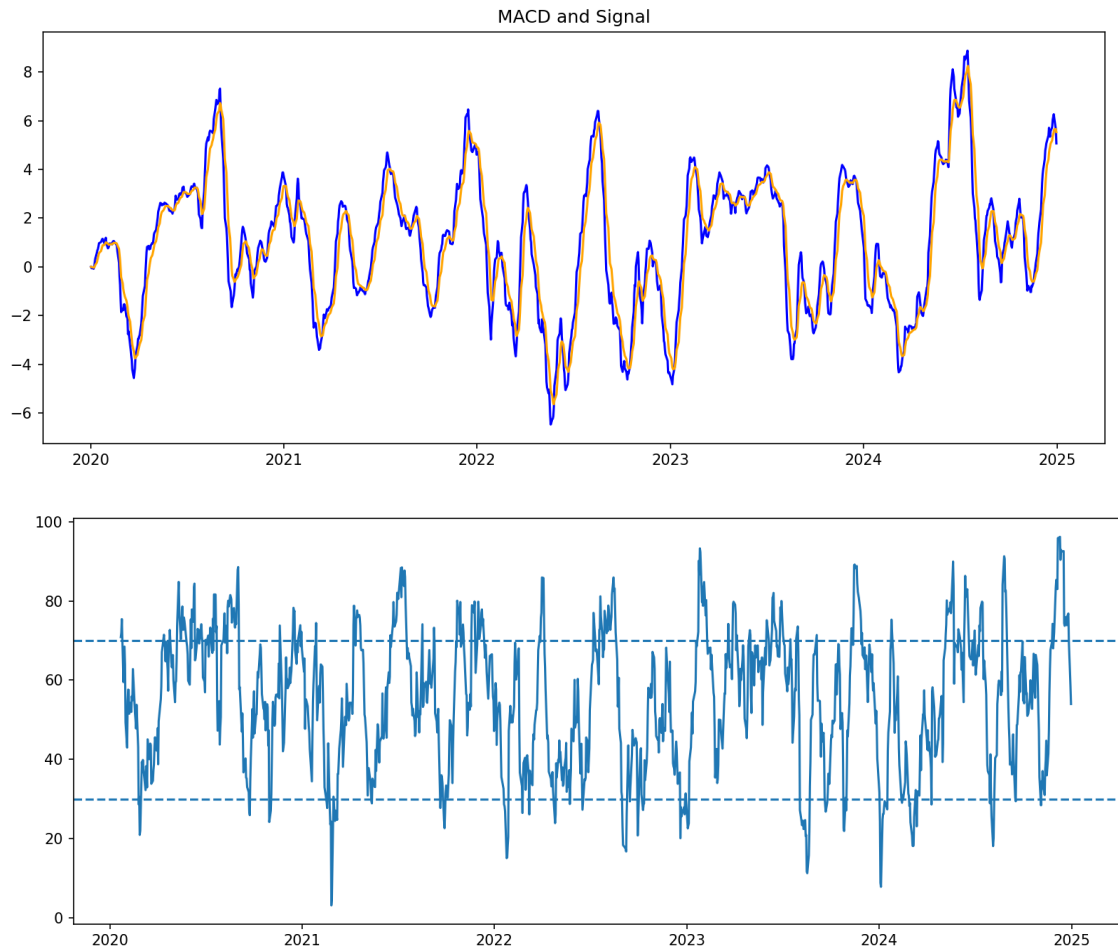
- **RSI (Relative Strength Index)**

Measures:

1. Overbought conditions
2. Oversold conditions

Range:

1. Above 70 → Overbought
2. Below 30 → Oversold



Data Preprocessing

Selected Features

- Close
- MACD
- RSI

Preprocessing Steps:

- 1.Remove missing values
- 2.Normalize using MinMaxScaler
- 3.Create time-series windows

Past 60 Days



LSTM Model



Next 15 Days

LSTM Architecture

Model Structure

Input (60×3)

↓

LSTM Layer (64 Units)

↓

Dropout (20%)

↓

LSTM Layer (32 Units)

↓

Dropout (20%)

↓

Dense Layer (32)

↓

Output Layer (15 Values)

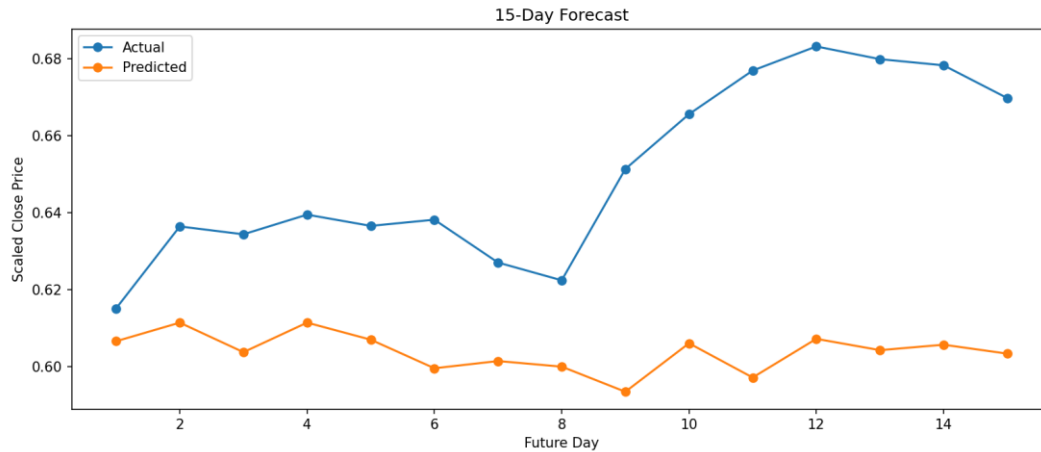
Training Parameters

- Optimizer – Adam
- Loss- MSE
- Epochs-30
- Batch size-32

Results

15-Day Forecast Graph

(the Actual vs Predicted graph)



Performance Metrics

RMSE = 0.07913076554136701

R^2 Score = 0.5727281625383858

Limitations

Challenges

- Stock market is highly volatile
- News events are not included
- Economic factors are not included
- Predictions depend only on historical data

Conclusion

Downloaded real stock data

Calculated MACD and RSI

Built LSTM forecasting model

Predicted next 15 days

Evaluated using RMSE and R^2

Future Improvements

1. More Technical Indicators

2. Hyperparameter Tuning
3. Testing on Multiple Stocks